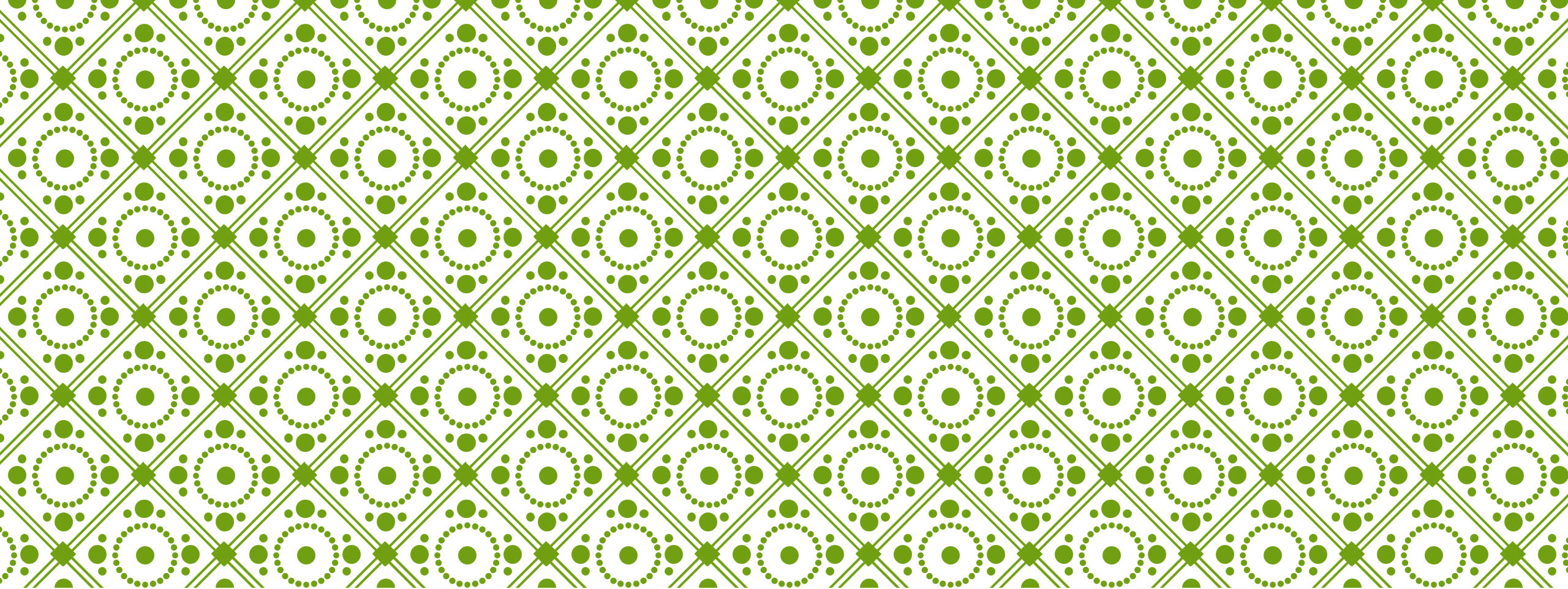




AI2002 — ARTIFICIAL INTELLIGENCE

SPRING 2024 - LECTURE 01-03

Presented By: Mr. Sandesh
Kumar

Slides taken from UC Berkley's
Artificial Intelligence Course

COURSE OUTLINE (1/3)

- ❖ Week – 1: Introduction to AI, Foundations of AI, History of AI, AI in Industry, the concept of neurons and neural networks, Basic components of AI, Identifying AI Systems, Branches of AI, Intelligent Agents, Agents and Environments, Sensors and Actuators.
- ❖ Week – 2: The Concept of Rationality, Performance Measures, Rationality, Rationality V/S Omniscience, The nature of the environment, Performance, Environment, Actuators, and Sensors (PEAS), Agent types, Properties of environments, The structure of Agents, Problem Representation: Introduction to Trees and Graphs.
- ❖ Week – 3: Problem-Solving by Searching: Problem-solving agents, Components of the problem, Formulating problems, Searching for solutions, Measuring problem-solving performance, Uniformed Searching.
- ❖ Week – 4: Informed heuristic search strategies, Local searching, Constraint Satisfaction Problems.

COURSE OUTLINE (2/3)

- ❖ Week – 5: Constraint Satisfaction Problems, Backtracking search for CSPs, Local search constraint satisfaction problems, the structure of problems.
- ❖ Week – 6: Mid – 1 Exam.
- ❖ Week – 7: Adversarial Search, Games, Optimal decisions in Games, the minimax algorithm, the alpha-beta pruning, Logical agents, First Order Logic, Inference in First-Order Logic.
- ❖ Week – 8: Uncertainty, acting under uncertainty, Uncertainty in AI, Fuzzy Logic Basic Probability, Conditional Probability, Bayes Rule, Probabilistic Reasoning, Representing knowledge in an uncertain domain, Bayesian Networks
- ❖ Week – 9: Dynamic Bayesian Network, Introduction to Model-driven and Data-driven approach and Review on probability and Linear Algebra, Performance evaluation [ROC confusion matrix], Supervised Learning Methods: Linear Regression and Logistic Regression
- ❖ Week – 10: Non-parametric Methods, Decision Trees, Neural Networks, Units in NN, Single Layer Feed-Forward and Multi-Layer Feed-Forward NN, Unsupervised Learning Methods: Clustering (K-Mean).

COURSE OUTLINE (3/3)

- Week – 11: Mid-2 Exam
- Week – 12: Reinforcement Learning: Agent, environment, reward, state, policy, Q value, model of the environment, Reinforcement learning algorithms.
- Week – 13: Reinforcement Learning: Type of Reinforcement Learning, Popular models of RL (Q and Markov decision process).
- Week – 14: Recent Trends in AI and applications of AI algorithms: Trends, Case Study of AI systems [Anomaly Detection].
- Week – 15: Revision
- Week – 16: Final Exam

COURSE MARKS DISTRIBUTION

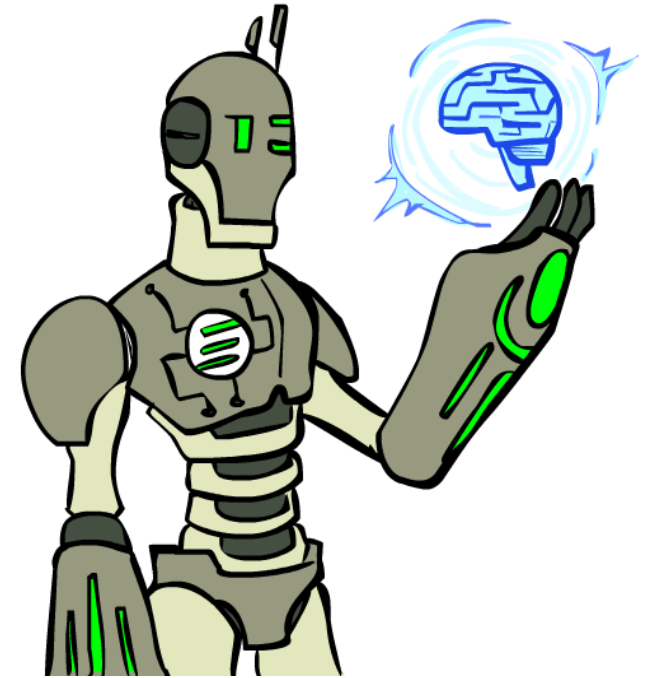
- Assignments: 4 and Weightage: 10
- Midterm Exams: 2 and Weightage: 15 each
- Project: 1 and Weightage: 10
- Final Exam: 1 and Weightage: 50

RECOMMENDED BOOKS AND REFERENCE MATERIAL

- Stuart Russel and Peter Norvig, *Artificial Intelligence, A Modern Approach*, 3rd Edition, Prentice Hall, Inc. 2010.

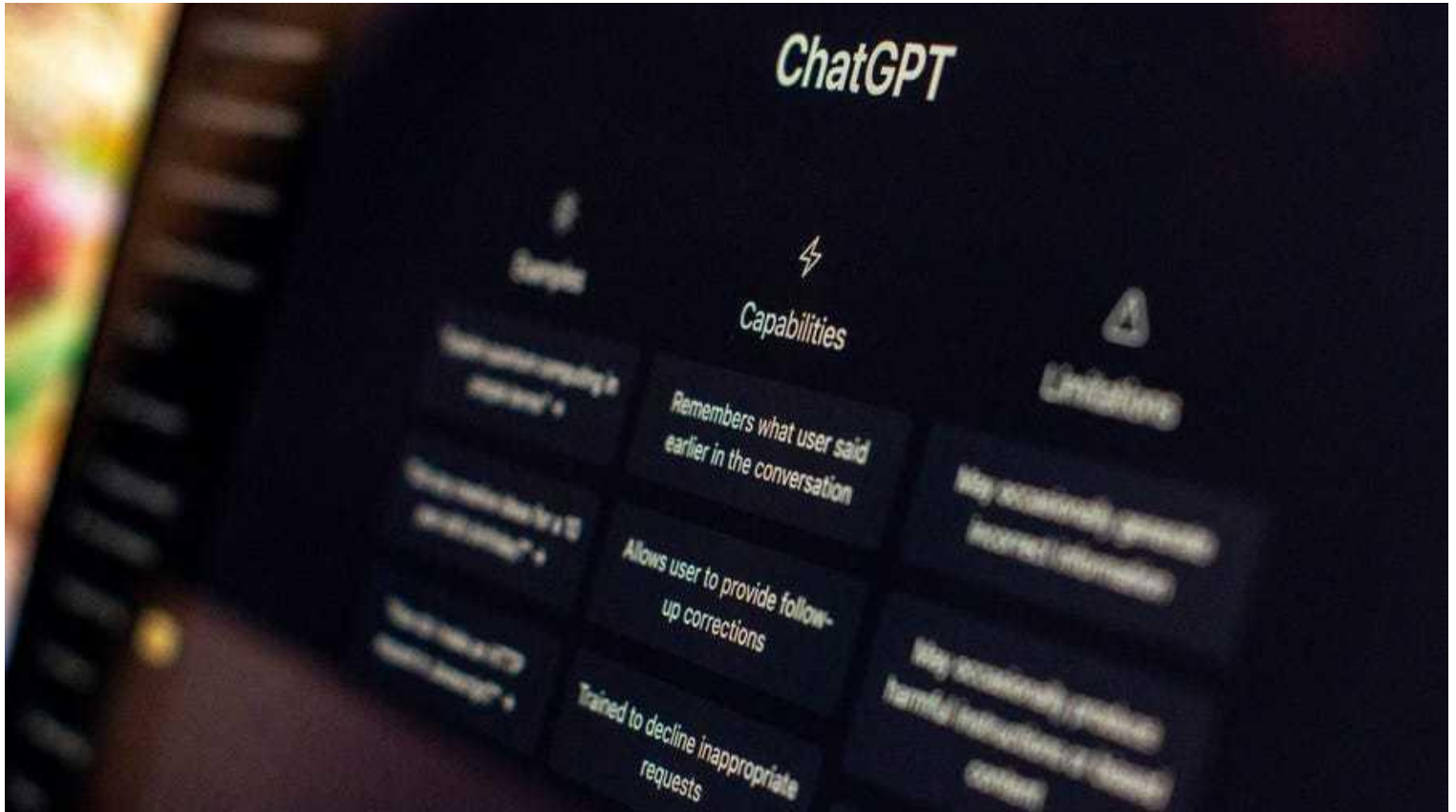
What is AI?

- What is artificial intelligence?
- What can AI do?
- What is this course?



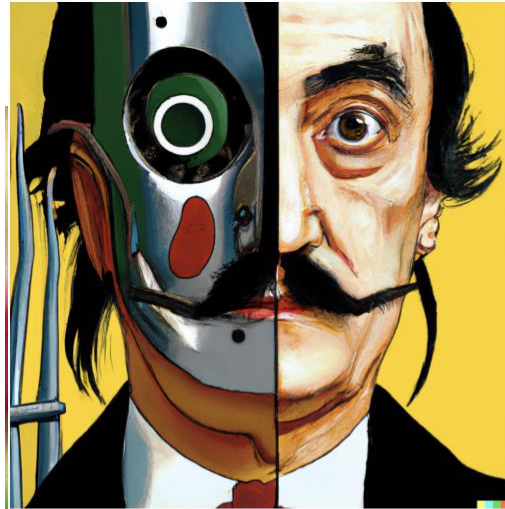
AI is having real-world impact

- Public imagination
 - Text assistants



AI is having real-world impact

- Public imagination
 - Text assistants
 - Image generation



vibrant portrait painting of Salvador Dalí with a robotic half face



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it



an espresso machine that makes coffee from human souls, artstation



panda mad scientist mixing sparkling chemicals, artstation

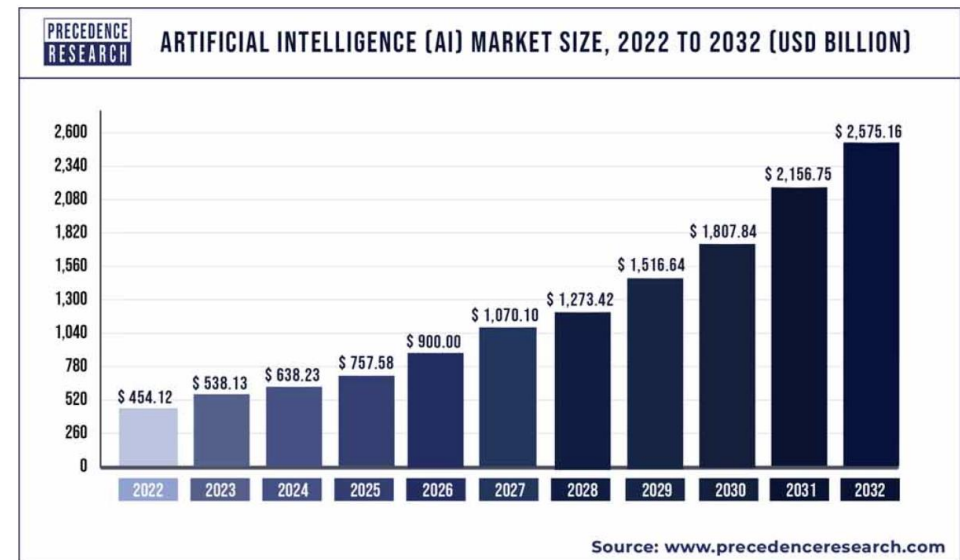


a corgi's head depicted as an explosion of a nebula

AI is having real-world impact

- Public imagination
- Economy
 - 454 billion USD globally

The global artificial intelligence (AI) market size was valued at USD 454.12 billion in 2022 and is expected to hit around USD 2,575.16 billion by 2032, progressing with a CAGR of 19% from 2023 to 2032. The North America artificial intelligence market was valued at USD 167.30 billion in 2022.



<https://www.precedenceresearch.com/artificial-intelligence-market>

AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

- Public imagination
- Economy
- Politics



AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law

Aug. 18, 2023, 12:18 PM; Updated: Aug. 18, 2023, 12:48 PM

AI-Generated Art Lacks Copyright Protection, D.C. Court Says (1)



Riddhi Setty
Reporter



Isaiah Poritz
Legal Reporter



Bloomberg Law, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor

Finance & economics | Free exchange

New research shows the robots are coming for jobs—but stealthily

Look beneath the aggregate economic numbers, and change is afoot

The Economist, 2021

The Optimist's Guide to Artificial Intelligence and Work

The focus of much discussion is on how it will replace jobs, but nothing is inevitable.

New York Times, 2023

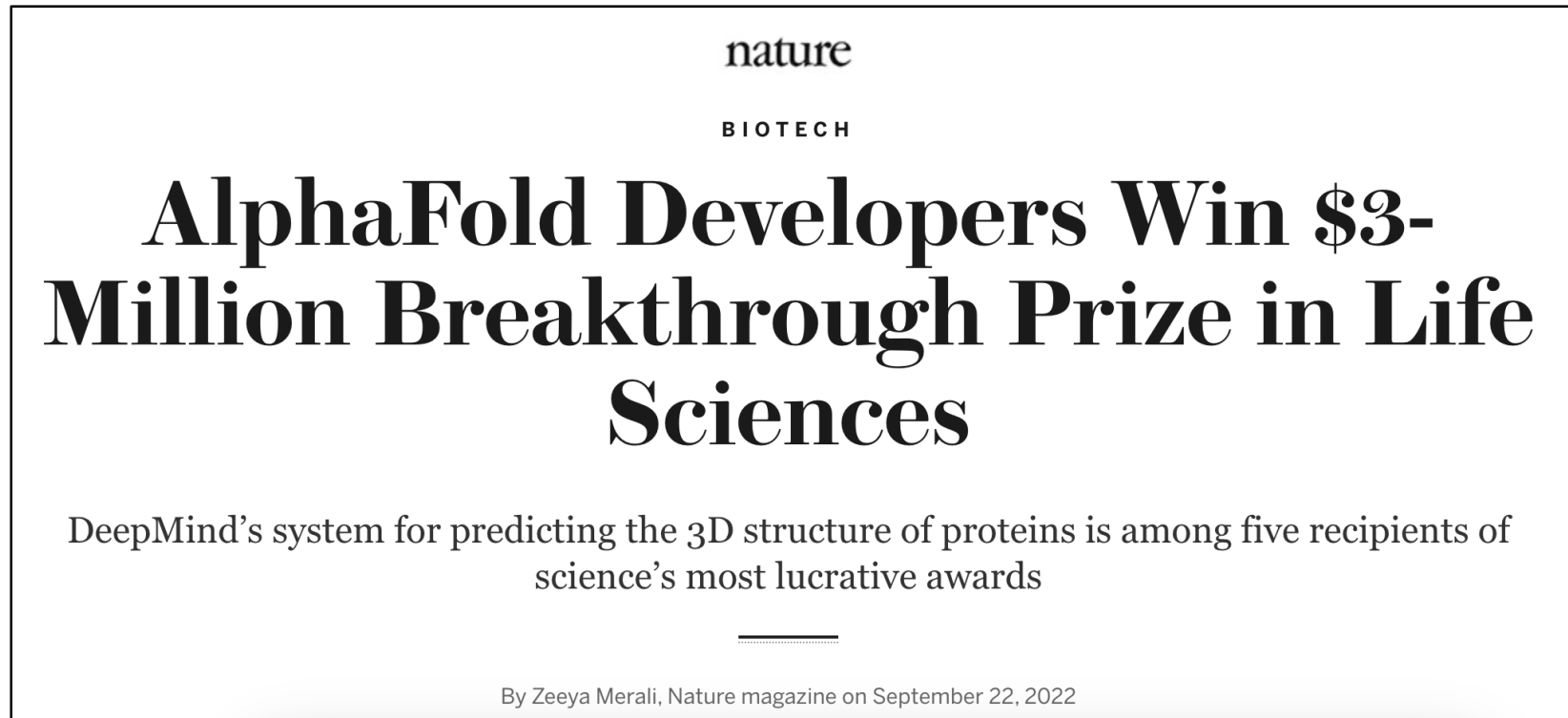
The human labor behind AI chatbots and other smart tools

Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences



Nature, 2022

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences



Wired, 2022

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences
- Education

BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

Follow

 2

Jan 18, 2023, 02:31pm EST

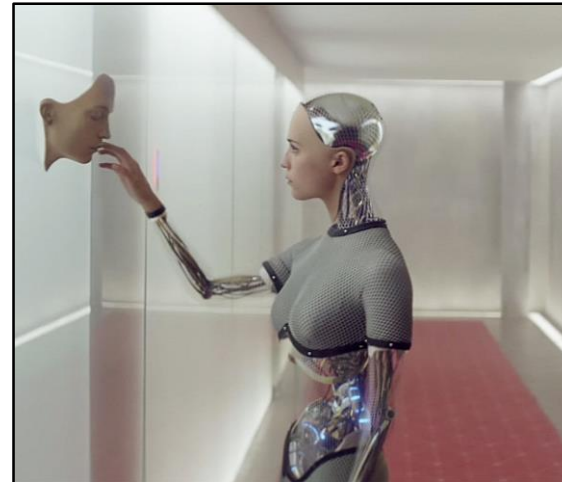
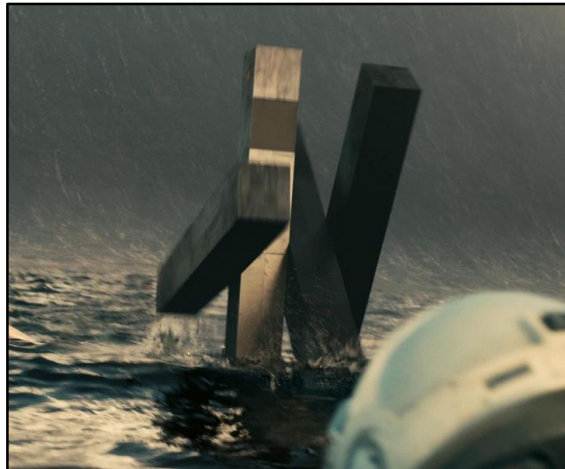
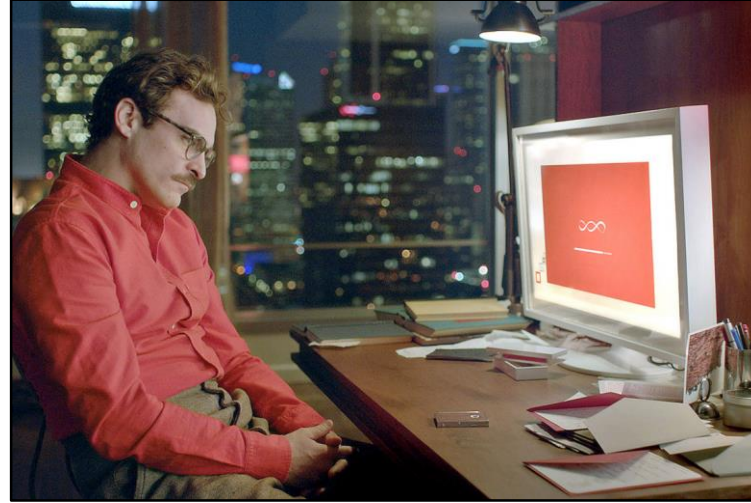
Forbes, 2023

AI is having real-world impact

- Public imagination
- Economy
- Politics
- Law
- Labor
- Sciences
- Education

Ok, but what actually is AI???

Science fiction AI?

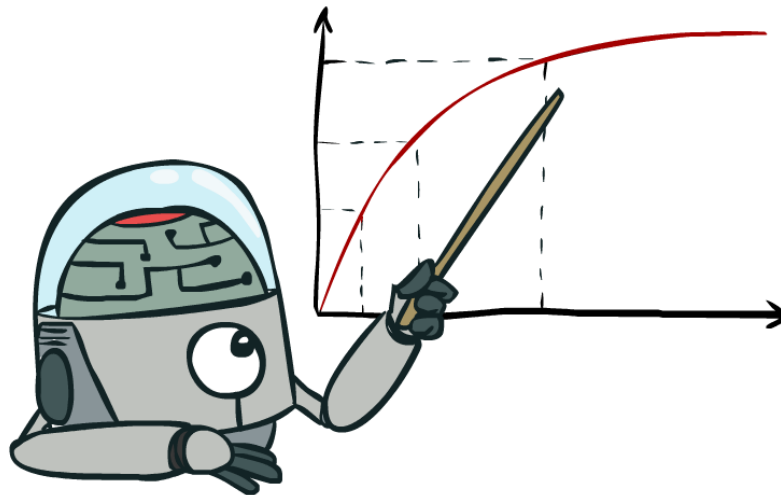


WHAT IS AI?

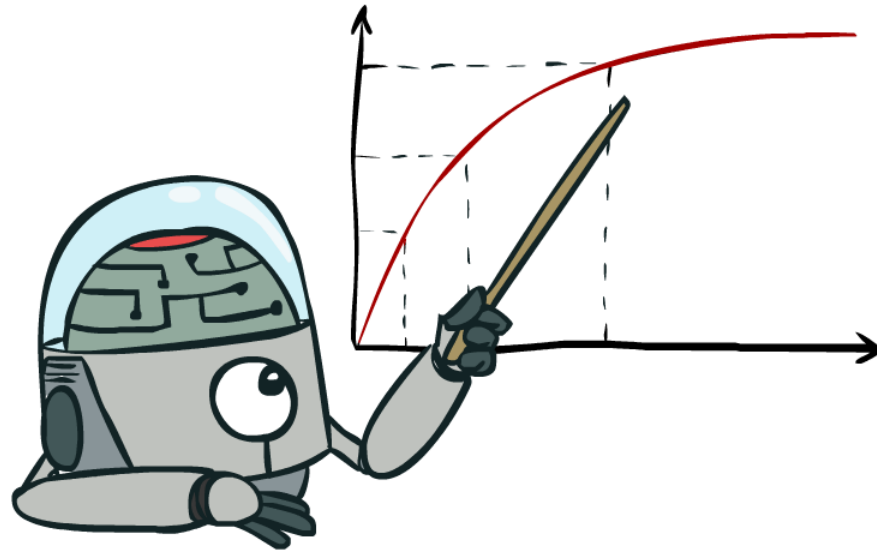
The science of making machines that:

Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
 - Rational: *maximally achieving pre-defined goals*
 - Goals are expressed in terms of the **utility** of outcomes
 - World is uncertain, so we'll use **expected** utility
 - Being rational means acting to **maximize your expected utility**



Maximize Your Expected Utility

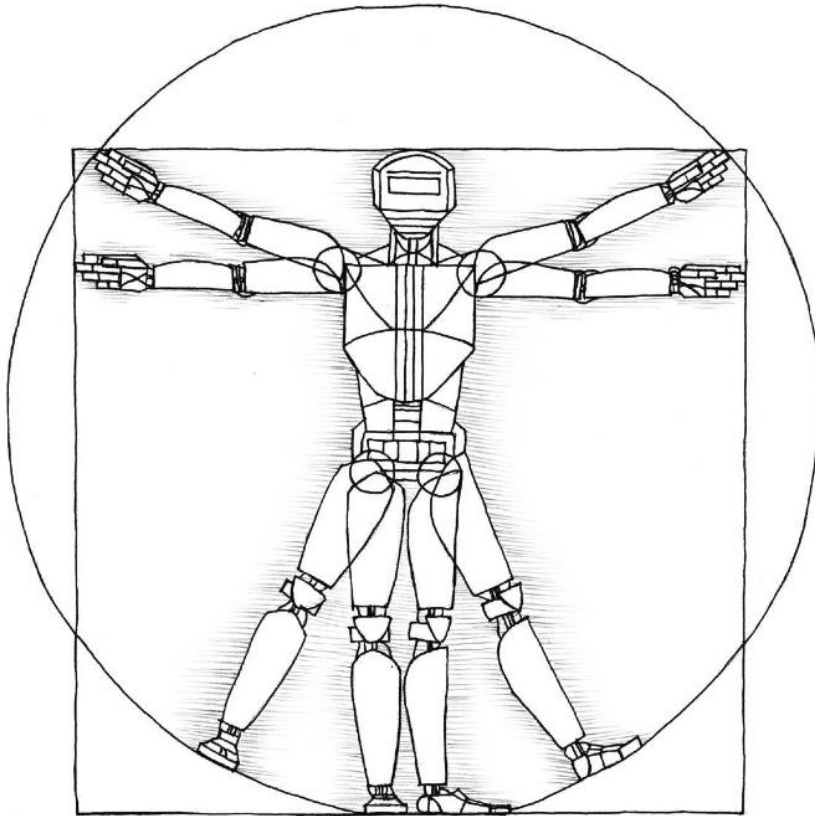


Perspectives on Intelligence

- Skills-based perspective
- “A system is only intelligent if it can do [X].”
 - Play chess?
 - Learn from experience?
 - Use words properly?
 - Make mistakes?
 - Not make mistakes?

Perspectives on Intelligence

- Embodiment perspective (Rodney Brooks)



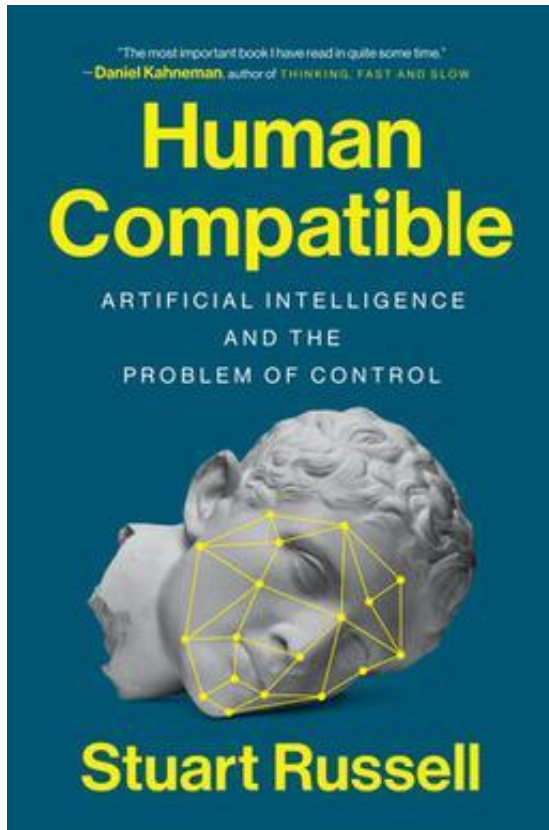
Perspectives on Intelligence

- Psychometrics perspective (François Chollet)
- “Measuring abilities, not skills [...] across a broad range of tasks, including tasks that were previously unknown to the ability-enabled system and its developers.”



Perspectives on Intelligence

- Human-compatible perspective (Stuart Russell)



1. Machine's objective is to maximize human utility.
2. Initially uncertain about human preferences.
3. Must learn about preferences from human behavior.

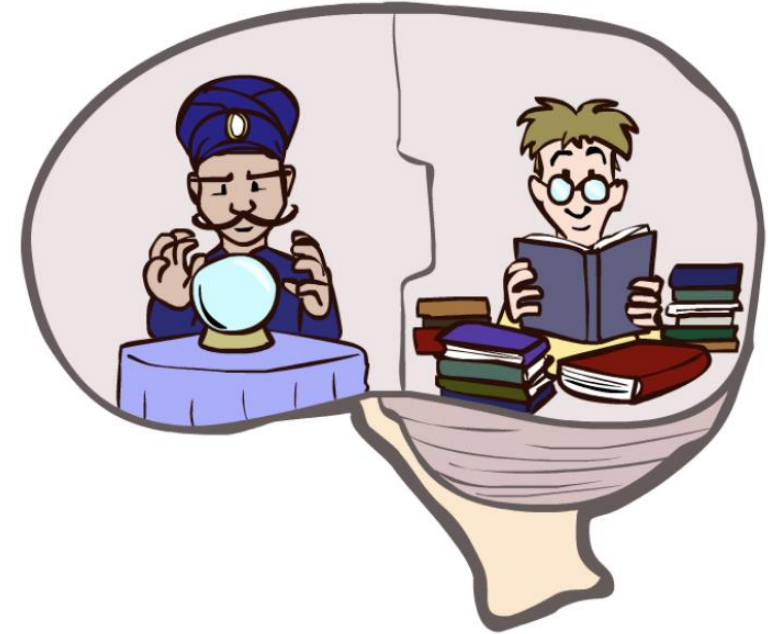
Perspectives on Intelligence

A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.

—Robert A. Heinlein

What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks
- *“Brains are to intelligence as wings are to flight”*
- We can't yet build AI on the scale of the brain
 - ~100T synapses in the human brain vs ~1.8T weights in GPT4
- Still, the brain can be a great inspiration for AI!



AI's official birth: Dartmouth, 1956



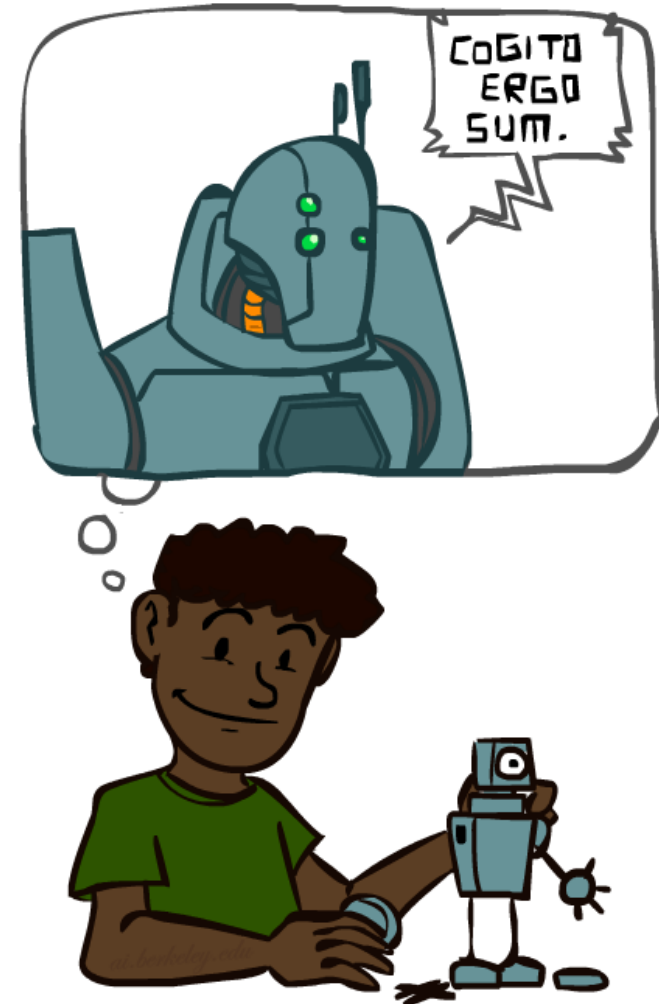
“An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer.***”

**John McCarthy and Claude Shannon
Dartmouth Workshop Proposal**

A (Short) History of AI

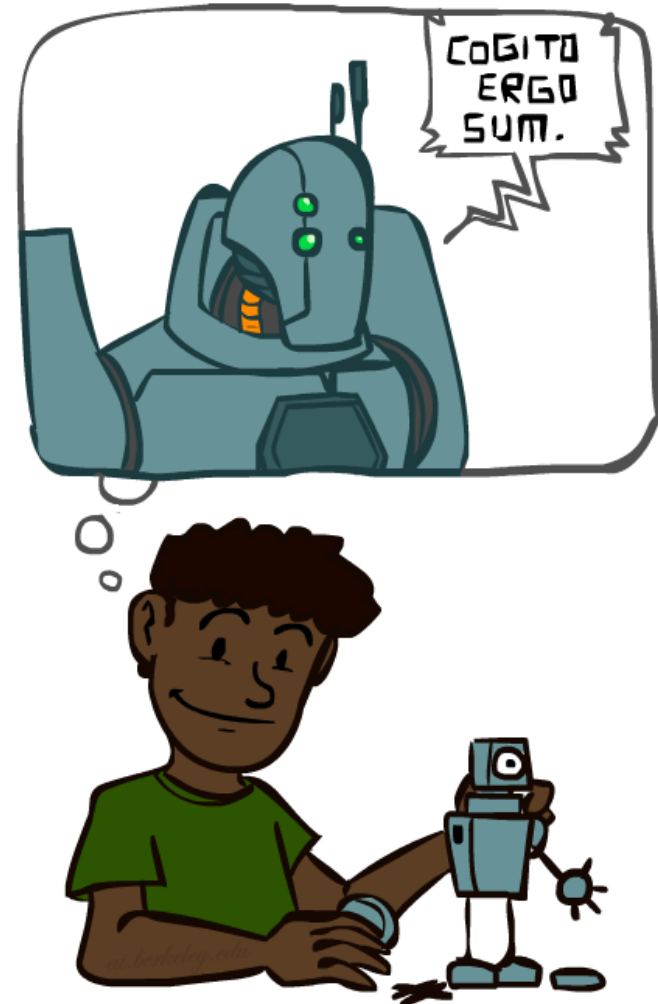
- 1940-1950: Early days: neural and computer science meet
 - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
 - 1950: Turing's “Computing Machinery and Intelligence”
- 1950—70: Excitement! Logic-driven
 - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
 - 1956: Dartmouth meeting: “Artificial Intelligence” adopted

*“We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning or any other feature of intelligence** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”*



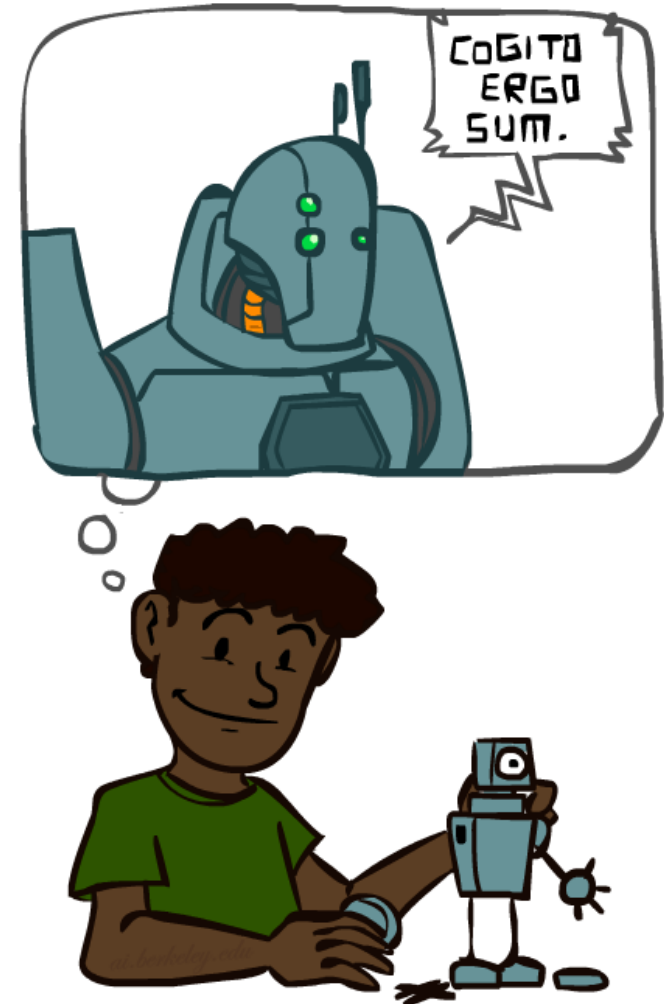
A (Short) History of AI

- **1940-1950: Early days: neural and computer science meet**
 - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
 - 1950: Turing's “Computing Machinery and Intelligence”
- **1950—70: Excitement! Logic-driven**
 - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
 - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
 - 1969: Minsky & Papert: perceptrons can't learn XOR/parity!
- **1970—90: Knowledge-based approaches**
 - 1969—79: Early development of knowledge-based systems
 - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
 - 1988—93: Expert systems industry busts: “AI Winter”
- **1990—2010: Statistical approaches, agents**
 - Resurgence of probability, focus on uncertainty
 - Agents and learning systems... “AI Spring”?
 - 1992: TD-Gammon achieves human-level play at backgammon
 - 1997: Deep Blue defeats Gary Kasparov at chess
 - 2002: Embodied AI; Roomba vacuum invented



A (Short) History of AI

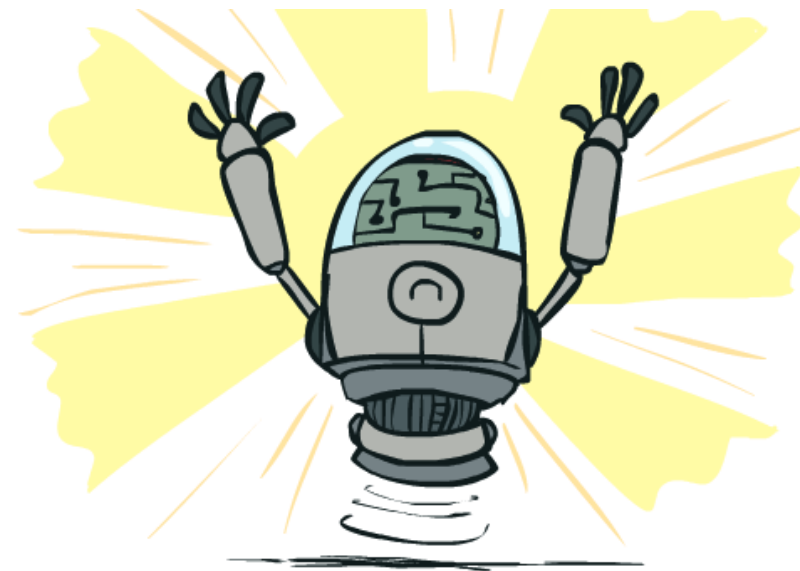
- 2010—2017: Big Data, GPUs, Deep Learning
 - 2011: Apple releases Siri
 - 2012: AlexNet wins ImageNet competition
 - 2015: DeepMind achieves “human-level” control in Atari games
 - 2016: DeepMind’s AlphaGo defeats Lee Sedol at Go
 - 2016: Google Translate migrates to neural networks
- 2017—: Scaling Up, Large Language Models
 - 2017: Google invents Transformer architecture
 - 2017: DeepStack/Libratus defeat humans at poker
 - 2018-2020: AlphaFold predicts protein structure from amino acids
 - 2021-2022: Modern text-to-image generation
 - 2022: OpenAI releases ChatGPT
 - 2023: Every other company also releases a chatbot



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- Play a decent game of table tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Berkeley Bowl?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- Play a decent game of table tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Berkeley Bowl?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?

Google's AlphaGo beats Lee Sedol at Go in 2016



But ... a plot twist in 2023!

DEEP BLUE WAS JUST THE START —

Man beats machine at Go in human victory over AI

Amateur exploited weakness in systems that have otherwise dominated grandmasters.

RICHARD WATERS, FINANCIAL TIMES - 2/19/2023, 4:51 AM

What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of table tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Berkeley Bowl?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of table tennis?
- ✗ Unload any dishwasher in any home?
- ✓ Drive safely along the highway?
- ? Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Berkeley Bowl?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?

California just opened the floodgates for self-driving cars

After a state regulatory board ruling, San Francisco will have 24/7 robotaxis on its streets. A Washington Post analysis shows how it's a pivotal moment for the industry.

By [Trisha Thadani](#) and [Jeremy B. Merrill](#)

Updated August 10, 2023 at 10:06 p.m. EDT | Published August 10, 2023 at 9:15 p.m. EDT

The Register

California DMV hits brakes on Cruise's SF driverless fleet after series of fender benders

50% chop effective 'immediately' as department investigates traffic and safety issues

 [Jude Karabus](#)

Mon 21 Aug 2023 // 14:28 UTC

What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of table tennis?
- ✗ Unload any dishwasher in any home?
- ✓ Drive safely along the highway?
- ? Drive safely along streets of San Francisco?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ? Discover and prove a new mathematical theorem?
- ✗ Perform a surgical operation?
- ✓ Translate spoken Chinese into spoken English in real time?
- ✓ Win an art competition?
- Write an intentionally funny story?
- Construct a building?



What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of table tennis?
- ✗ Unload any dishwasher in any home?
- ✓ Drive safely along the highway?
- ? Drive safely along streets of San Francisco?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ? Discover and prove a new mathematical theorem?
- ✗ Perform a surgical operation?
- ✓ Translate spoken Chinese into spoken English in real time?
- ✓ Win an art competition?
- ? Write an intentionally funny story?
- Construct a building?



You

Please write me a very funny, and extremely short story.

< 6 / 6 >



ChatGPT

Once there was a chicken who was terribly afraid of crossing roads. One day, it mustered up all its courage, stepped onto the road, and halfway across, it suddenly stopped and said, "Wait, why am I doing this again?"

< 12 / 12 >    

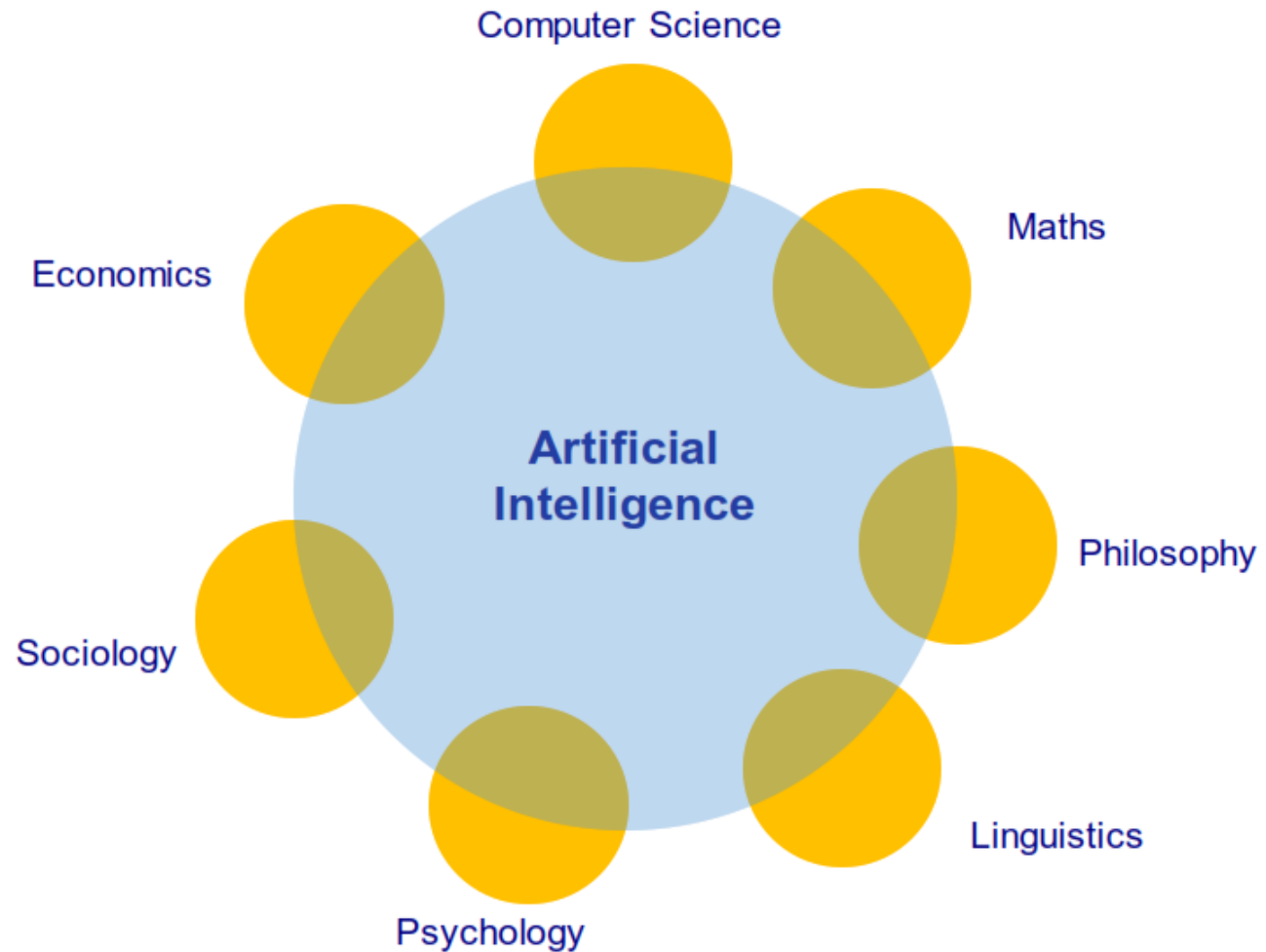
What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of table tennis?
- ✗ Unload any dishwasher in any home?
- ✓ Drive safely along the highway?
- ? Drive safely along streets of San Francisco?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ? Discover and prove a new mathematical theorem?
- ✗ Perform a surgical operation?
- ✓ Translate spoken Chinese into spoken English in real time?
- ✓ Win an art competition?
- ? Write an intentionally funny story?
- ✗ Construct a building?

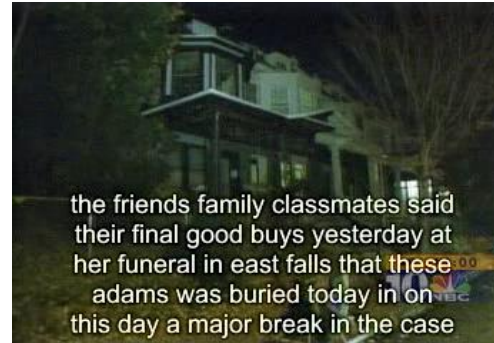


FOUNDATIONS OF AI



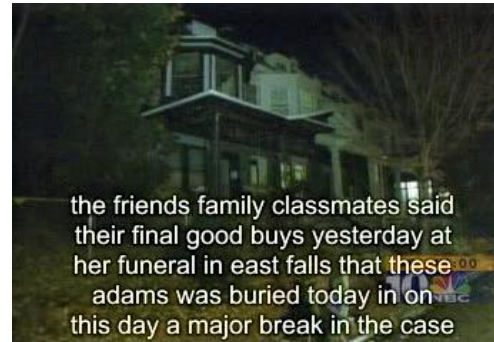
Natural Language

- Speech technologies (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems



Natural Language

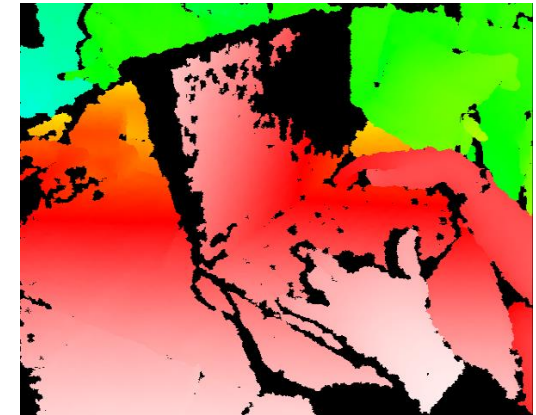
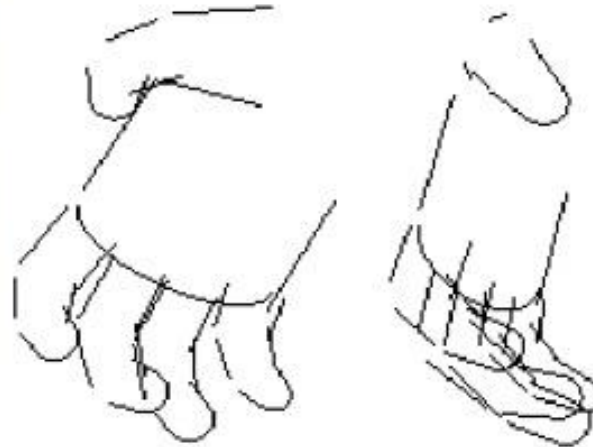
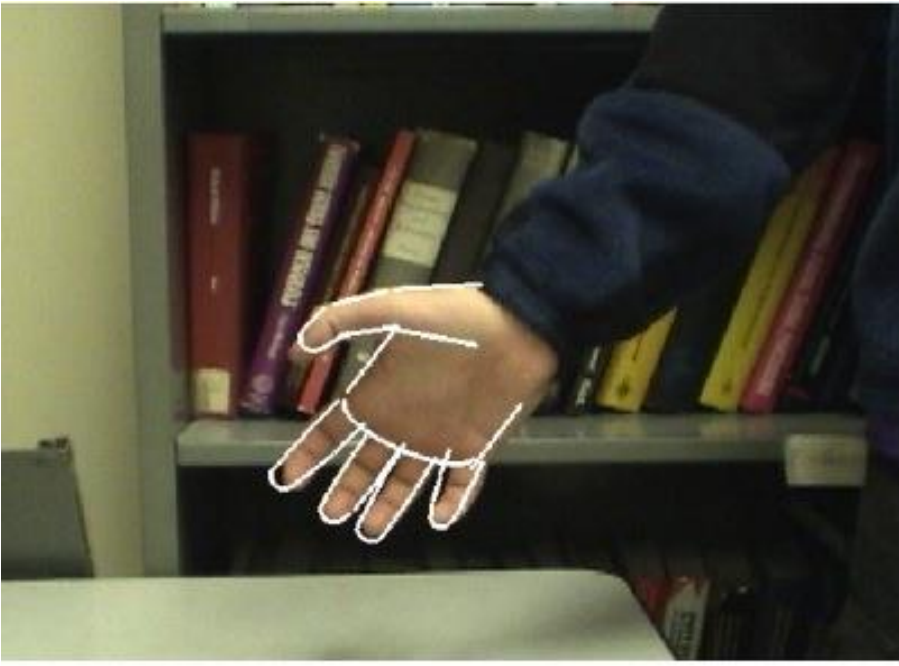
- Speech technologies (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems
- Language processing technologies
 - Question answering
 - Machine translation



- Web search
- Text classification, spam filtering, etc...

Vision (Perception)

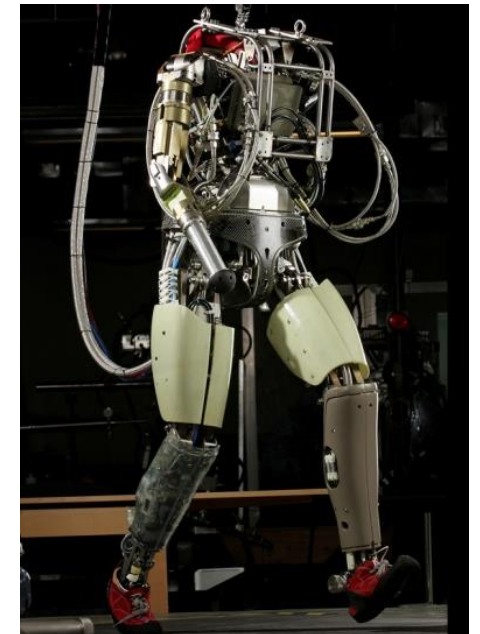
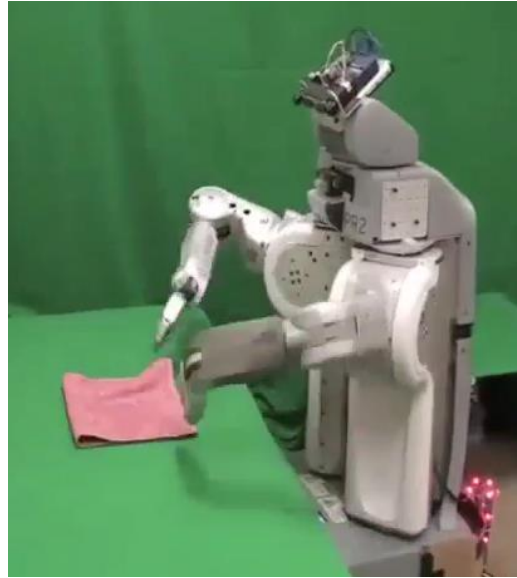
- Object and face recognition
- Scene segmentation
- Image classification



Images from Erik Sudderth (left), wikipedia (right)

Robotics

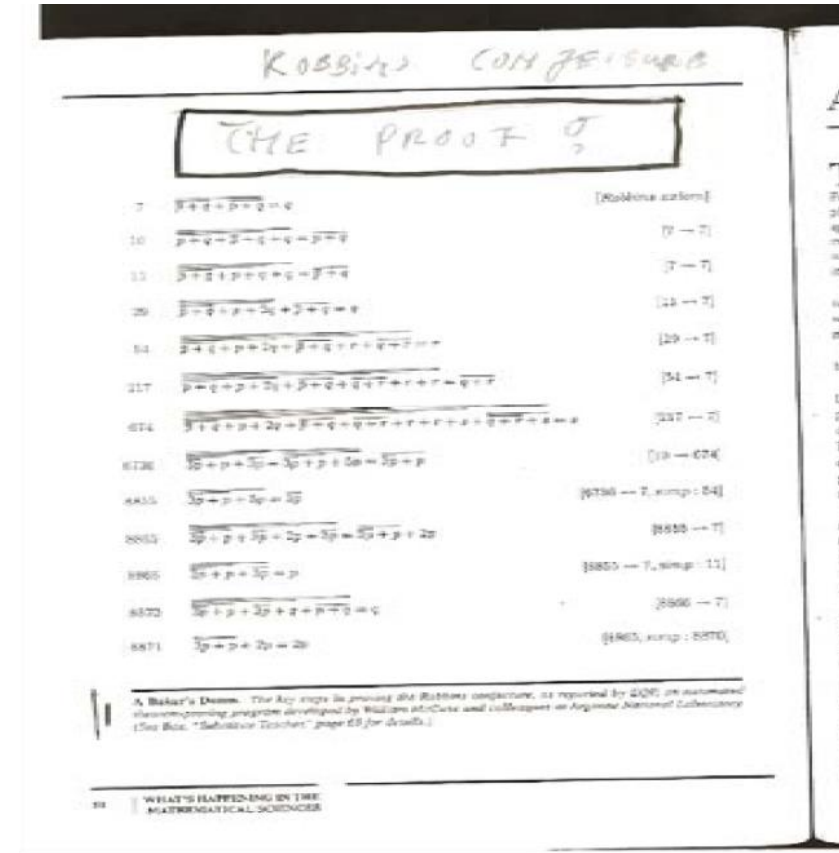
- Robotics
 - Part mech. eng.
 - Part AI
 - Reality much harder than simulations!
- Technologies
 - Vehicles
 - Rescue
 - Soccer!
 - Lots of automation...
- In this class:
 - We ignore mechanical aspects
 - Methods for planning
 - Methods for control



Images from UC Berkeley, Boston Dynamics, RoboCup, Google

Logic

- Logical systems
 - Theorem provers
 - NASA fault diagnosis
 - Question answering
- Methods:
 - Deduction systems
 - Constraint satisfaction
 - Satisfiability solvers (huge advances!)



Game Playing

- **Classic Moment: May, '97: Deep Blue vs. Kasparov**
 - First match won against world champion
 - “Intelligent creative” play
 - 200 million board positions per second
 - Humans understood 99.9 of Deep Blue's moves
 - Can do about the same now with a PC cluster
- **Open question:**
 - How does human cognition deal with the search space explosion of chess?
 - Or: how can humans compete with computers at all??
- **1996: Kasparov Beats Deep Blue**

“I could feel --- I could smell --- a new kind of intelligence across the table.”
- **1997: Deep Blue Beats Kasparov**

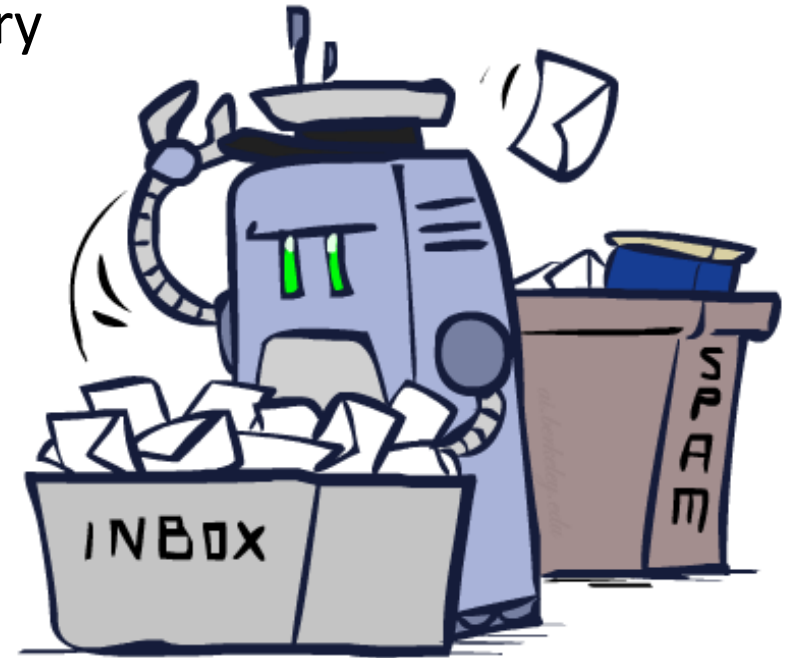
“Deep Blue hasn't proven anything.”
- **Huge game-playing advances recently, e.g. in Go!**



Decision Making

- Applied AI involves many kinds of automation

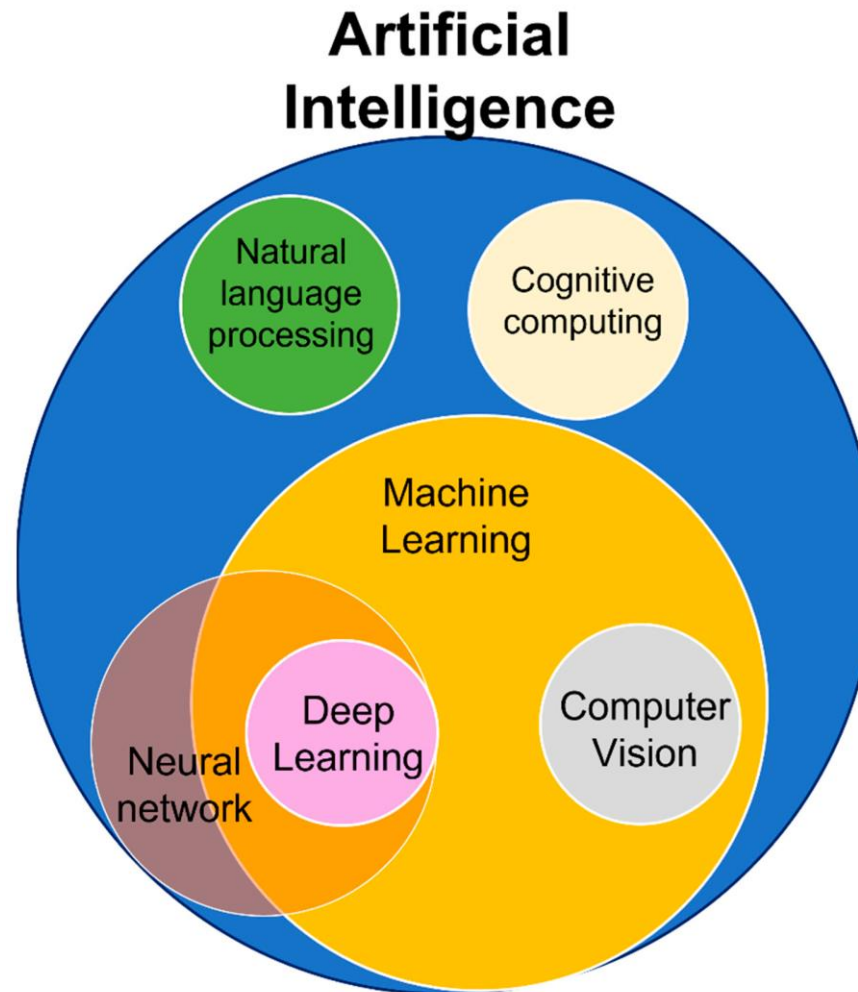
- Scheduling, e.g. airline routing, military
- Route planning, e.g. Google maps
- Medical diagnosis
- Web search engines
- Spam classifiers
- Automated help desks
- Fraud detection
- Product recommendations
- ... Lots more!



CONCEPT OF ARTIFICIAL NEURON AND NEURAL NETWORK

- Artificial neurons (also called Perceptron's, Units or Nodes) are the simplest elements or building blocks in a neural network. They are inspired by biological neurons that are found in the human brain.
- A neural network is a method in artificial intelligence that teaches computers to process data in a way that is inspired by the human brain. It is a type of machine learning process, called deep learning, that uses interconnected nodes or neurons in a layered structure that resembles the human brain.

BASIC COMPONENTS OF AI (1/2)



BASIC COMPONENTS OF AI (2/2)

- Machine learning: allows software applications to become more accurate at predicting outcomes without the requirement for explicit programming.
- Deep learning: a subset of ML that learns by processing data based on artificial neural networks with representation learning.
- Neural network: NN mimics the way human brain operates with a series of algorithms that endeavours to recognize underlying relationships in a set of data.
- Natural language processing (NLP) is a subfield of computer science and artificial intelligence concerned with the recognition, interpretation, and production of human language and speech.
- Computer vision: a subset of machine learning and pattern identification that enables computers to interpret image content based on graphs, tables, PDF pictures, and videos.
- Cognitive computing: this is related to technology platforms that attempt to mimic the way a human brain works.

IDENTIFYING STAGES OF AI (1/3)

- Artificial Narrow Intelligence: Also known as Weak AI, ANI is the stage of Artificial Intelligence involving machines that can perform only a narrowly defined set of specific tasks. At this stage, the machine does not possess any thinking ability, it just performs a set of specific tasks. Examples of weak AI include Siri, Alexa, Self-driving cars, Alpha-GO



IDENTIFYING AI STAGES AND TYPES (2/3)

- Artificial General Intelligence: Also known as Strong AI, AGI is the stage in the evolution of Artificial Intelligence wherein machines will possess the ability to think and make decisions just like us humans.



IDENTIFYING AI STAGES AND TYPES (3/3)

- Artificial Super Intelligence: it is the stage of Artificial Intelligence when the capability of computers will surpass human beings. ASI is currently a hypothetical situation as depicted in movies and science



TYPES OF AI (1/4)

- **Reactive Machine AI:** This type of AI includes machines that operate solely based on the present data, taking into account only the current situation. Reactive AI machines cannot form inferences from the data to evaluate their future actions. They can perform a narrowed range of pre-defined tasks. An example of Reactive AI is the famous IBM Chess program that beat the world champion, Garry Kasparov.



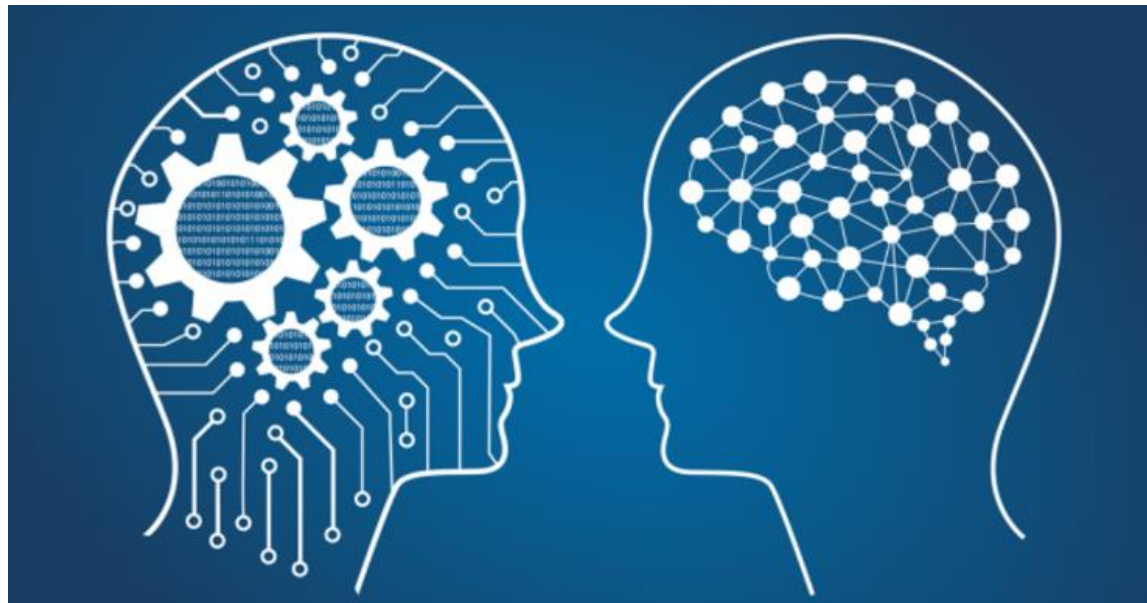
TYPES OF AI (2/4)

- Limited Memory AI: Like the name suggests Limited Memory AI, can make informed and improved decisions by studying the past data from its memory. Such an AI has a short-lived or a temporary memory that can be used to store past experiences and hence evaluate future actions. Self-driving cars are Limited Memory AI.



TYPES OF AI (3/4)

- Theory of Mind AI: It is a more advanced type of Artificial Intelligence. This category of machines is speculated to play a major role in psychology. This type of AI will focus mainly on emotional intelligence so that the human believes and thoughts can be better comprehended.



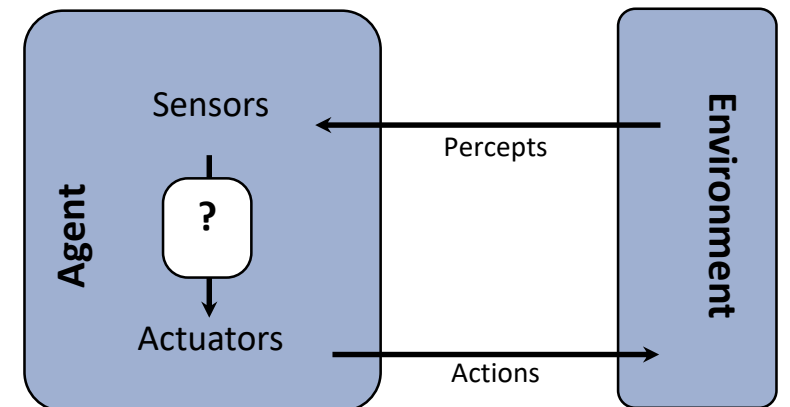
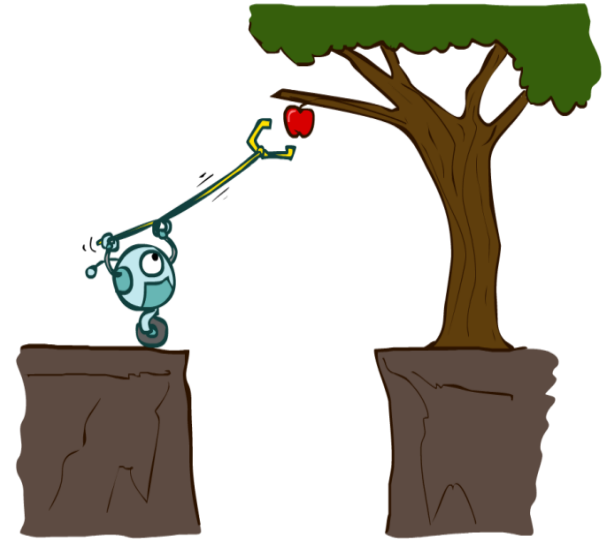
TYPES OF AI (4/4)

- Self-Aware AI: where machines have their own consciousness and become self-aware. This type of AI is a little far fetched given the present circumstances



Designing Rational Agents

- An **agent** is an entity that perceives and acts.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions
- This course is about:
 - General AI techniques for a variety of problem types
 - Learning to recognize when and how a new problem can be solved with an existing technique



Pac-Man as an Agent

